LESSON 10 · CHEAT SHEET

pandas II: Pivots & Lookups

COUNTING

```
q["region"].value_counts()      # COUNTIF, sorted
q["outcome"].value_counts(normalize=True)
q["monster"].nunique()          # distinct values
pd.crosstab(q["band"], q["outcome"])
pd.crosstab(a, b, normalize="index")  # % per row
```

GROUPBY = PIVOTTABLE

```
g = q.groupby("region")["reward"]  # rows, values
g.mean()   g.median()   g.sum()   g.count()
g.agg(["count", "mean", "median", "max"])
g.sum().sort_values(ascending=False)
g.mean().idxmax()             # which row is max

q.groupby(["monster", "outcome"])["quest_id"]
 .count().unstack(fill_value=0)  # 2nd key -> cols
q.groupby("class").agg(
    trips=("quest_id", "count"),
    loot=("loot_share", "sum"))
```

PIVOT_TABLE

```
q.pivot_table(index="region",    # Rows box
              columns="outcome", # Columns box
              values="quest_id", # Values box
              aggfunc="count",   # Summarise by
              margins=True,      # Grand Total
              fill_value=0)      # empty cells
# no aggfunc? it takes the MEAN
```

MERGE = XLOOKUP

```
q.merge(regions, on="region", how="left")
# left:  keep all of q, NaN if no match
# inner: keep matches only (rows vanish!)
# outer: keep everything from both
party.merge(heroes, on="hero_id", how="left")
len(q)   # check before AND after
```

MISSING VALUES (NAN)

```
m.isna().sum()               # per column
m["warden"].isna().sum()     # one column
m[m["warden"].isna()]        # the rows
m["warden"] = m["warden"].fillna("(none)")
m = m.dropna(subset=["warden"])
m["danger"].mean()           # ignores NaN
```

BANDS

```
pd.qcut(q["reward"], 4, labels=[...]) # same count
pd.cut(q["days"], [0, 5, 10, 99])    # fixed edges
q.groupby("band", observed=True)["reward"]
 .agg(["min", "max"])
```

PICKING CELLS FROM A TABLE

```
t = q.groupby("class").agg(...)
t.loc["Bard", "difficulty"]  # one cell
t["difficulty"]              # one column
t.loc["Bard"]                # one row
t.reset_index()              # index -> column
t.sum(axis=1)                # row totals
```

EXCEL → PANDAS

COUNTIF	`value_counts()`
PivotTable	`groupby("a")["v"].mean()`
rows + columns	`pivot_table(...)`
Grand Total	`margins=True`
XLOOKUP	`merge(how="left")`
#N/A	`NaN`
IFERROR(x, 0)	`.fillna(0)`
QUARTILE groups	`pd.qcut(col, 4)`

GOTCHAS

- Two keys: `groupby(["a", "b"])`, a list
- `how="inner"` drops unmatched rows silently; count rows
- `fillna` returns a new column: assign it back
- groupby drops NaN keys: fill them first
- Merging a row-per-hero table repeats each quest's reward